

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Concept Mapping

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 1 – Concept Mapping
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.</i>		
Student Name:	P. Harsha vardhan	Reg. No.:	192411179
Section / Batch:	CSE	Date:	08/07/2026

Code of Conduct

p. Harsha Vardhan/192411179 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should be with following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	2	20
Search Data Structure	Linear Search and Binary Search	3	20
Recursion, Performance analysis	Recursion	4	20
Tree Data Structures	Classifications of Tree data structure	5	20
GRAND TOTAL:		100 Marks	

Annexure A – Concept Mapping

1. A student attendance system records daily presence data. Construct a concept map linking Array → Fixed Size → Sequential Access → Linked List → Dynamic Allocation → Flexibility. Explain how each structure works and justify which is more suitable for dynamic data updates. (20 Marks)
2. A weather monitoring system collects temperature, humidity, and wind data continuously. Construct a concept map linking Primitive Data Types → Non-Primitive Structures → Linear Structures → Non-Linear Structures → Applications. Include examples and explain how each structure supports continuous data processing. (20 Marks)
3. A digital archive stores millions of documents. Construct a concept map linking Searching → Linear Search → Binary Search → Sorted Data → Time Complexity → Performance. Analyze the impact of data organization on search efficiency and justify the best technique. (20 Marks)
4. A program processes nested expressions using recursion and may encounter deep nesting. Construct a concept map linking Recursion → Call Stack → Stack Overflow → Error Handling → Efficiency. Analyze potential issues and justify strategies to handle them. (20 Marks)
5. A company maintains hierarchical employee records with multiple levels of management. Construct a concept map connecting Tree → Nodes → Levels → Traversal (DFS/BFS) → Searching → Efficiency. Analyze how traversal techniques affect performance and justify the most suitable approach for retrieving employee data. (20 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____

LINKED LIST VS ARRAY IN C

ARRAYS



- ✓ Fast random access (indexing)
- ✓ Easy to use, memory continuous
- ✗ Fixed size
- ✗ Expensive insert/delete in middle

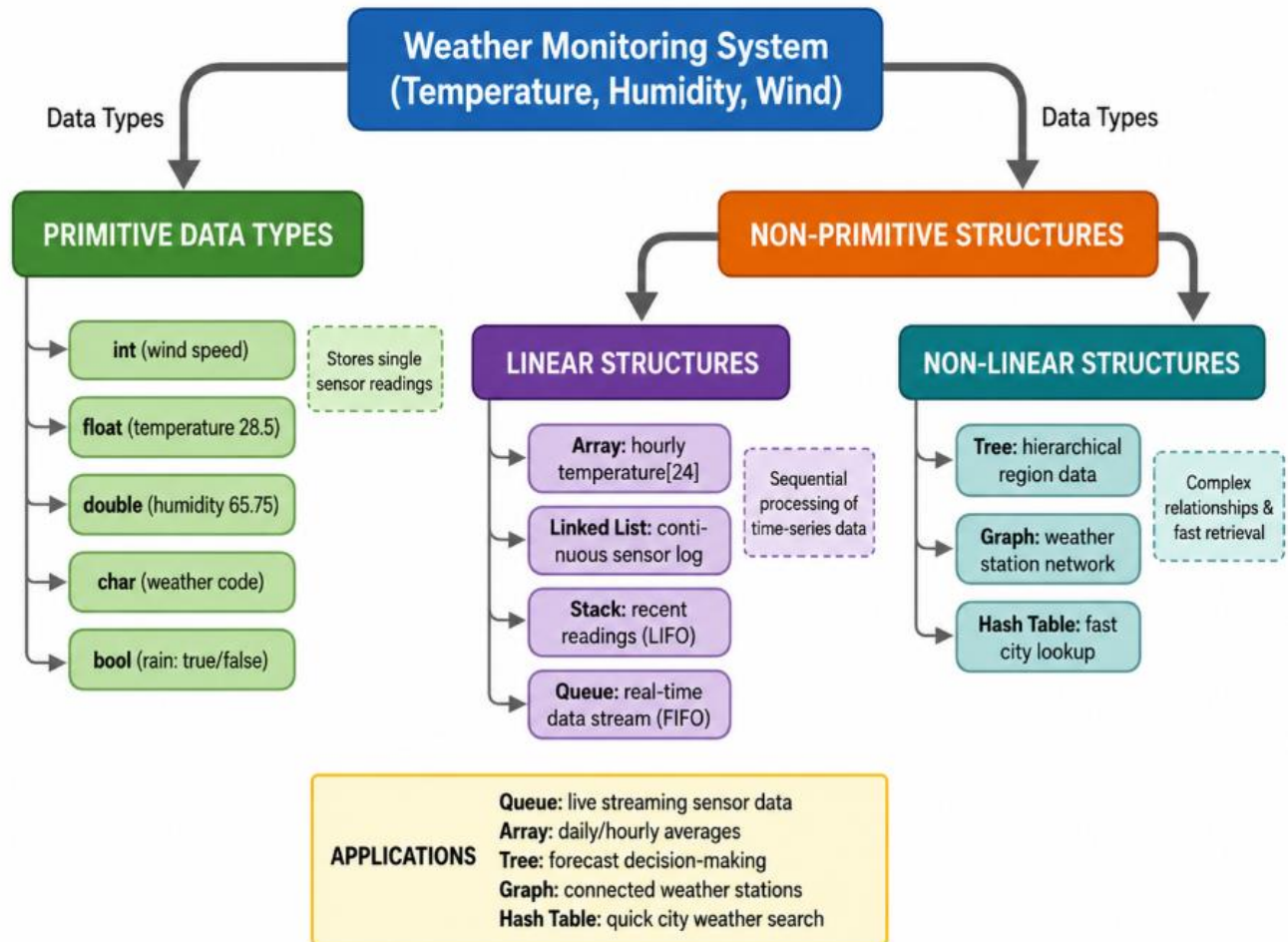
LINKED LISTS



- ✓ Dynamic size (grow/shrink easily)
- ✓ Efficient insert/delete (just update pointers)
- ✗ Slower access (sequential search)
- ✗ Extra memory for pointers

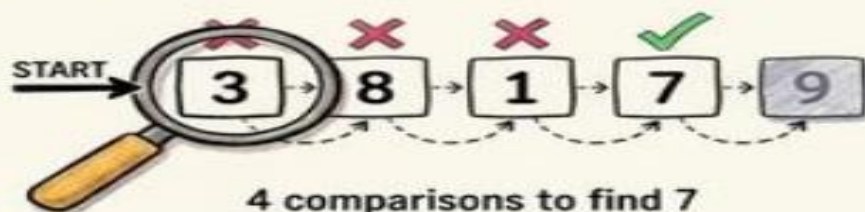
Linked Lists are useful when frequent insertions/deletions are needed.

2. WEATHER MONITORING SYSTEM - DATA STRUCTURES CONCEPT MAP



LINEAR **VS** BINARY SEARCH

LINEAR SEARCH



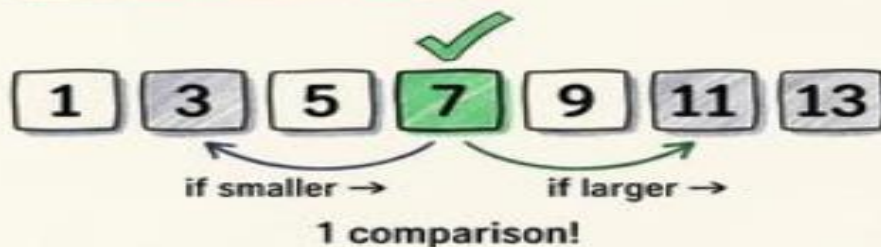
HOW IT WORKS

- 1 Start at the first item
- 2 Compare each item to the target
- 3 Stop when found — or reach the end

Works on: ANY list
(sorted or not)

Worst case: 1000 items
→ 1000 comparisons

BINARY SEARCH



HOW IT WORKS

- 1 Find the middle item
- 2 Match? Done.
- 3 Target smaller? Search left half.
- 4 Target larger? Search right half.

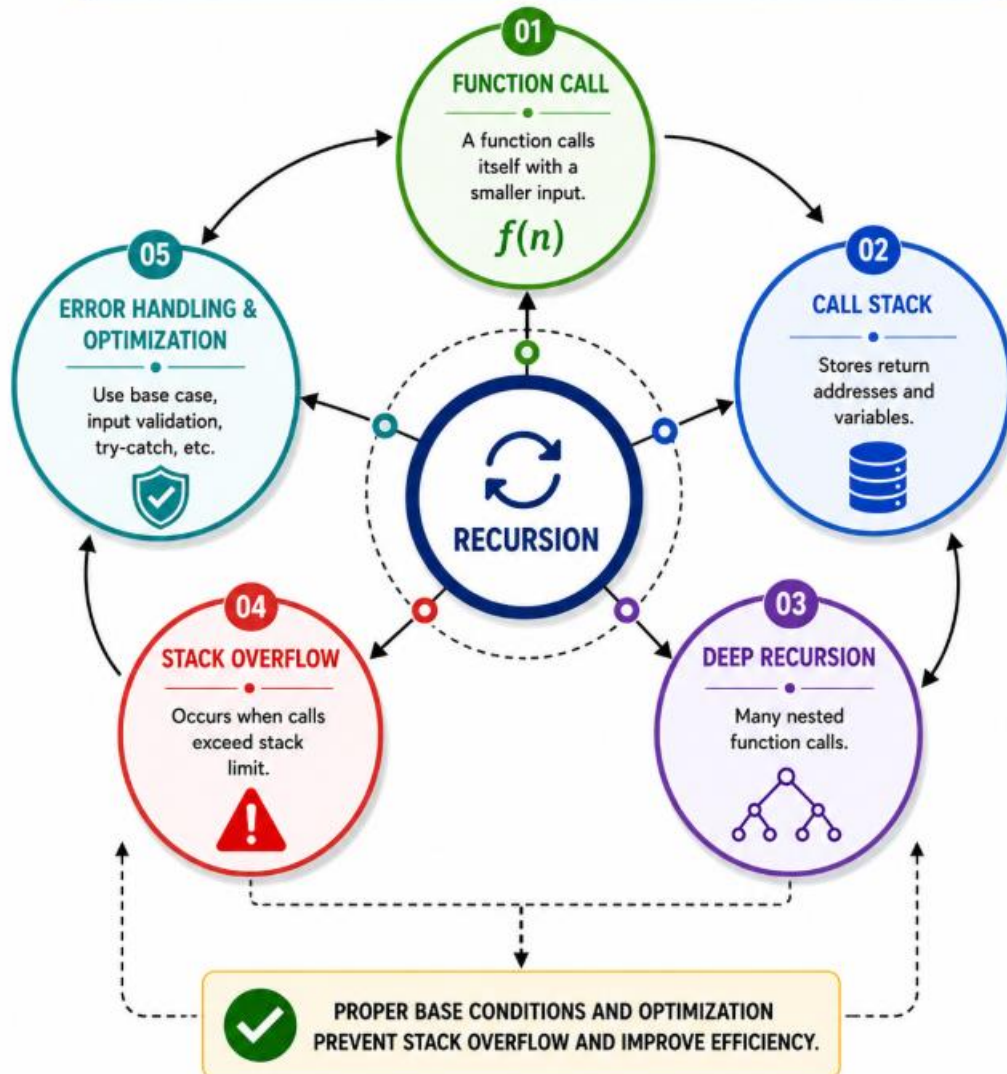
Works on: SORTED
lists only

Worst case: 1000 items
→ ~10 comparisons

	LINEAR	BINARY
Needs sorted list?	NO	YES
1000 items, worst case	1000	10

The Adaptive Toolkit

4. RECURSION, PERFORMANCE ANALYSIS



5. TREE DATA STRUCTURES – DETAILS



- Each node represents an employee.
- It contains information like ID, Name, Designation, etc.

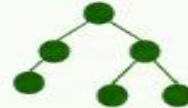


- Nodes are arranged in different levels.
- Level 0 is the top level (e.g., CEO).
- Lower levels represent subordinates.



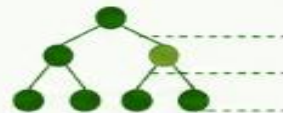
DFS (Depth First Search)

- Goes deep into a branch first.
- Types: Preorder, Inorder, Postorder.



BFS (Breadth First Search)

- Visits level by level.
- Uses a queue to explore all nodes at the current level first.



- Used to find a specific employee.
- Search by ID or Name.



- Performance depends on traversal choice.
- Affects time complexity and space used.



MOST SUITABLE APPROACH

- ✓ **DFS** – Best for searching a specific branch or detailed information
- ✓ **BFS** – Best for level-wise search and shortest level information

